

Virtual Firewall Lab on FortiGate-VM

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Environment:

1. **FortiGate-VM** (FortiOS 7.6.7) on VMware Workstation
2. **Windows-VM** (Internal)
3. **Ubuntu-VM** (Guest)

Overview:

A small office network built on FortiGate-VM, with two separate networks: one for employees and one for guests. Guests can reach the internet but cannot reach the employee network. Employee browsing is filtered to block social media and gambling sites. All of this was tested and confirmed through the firewall logs.

1. Network Segmentation

Divided the network into three separate zones using dedicated physical interfaces, with administrative access restricted per zone — HTTPS and SSH on WAN, HTTPS on Internal, and ICMP only on Guest.

- **Port 1 (WAN):** 192.168.117.10/24 — External connectivity
- **Port 2 (Internal):** 192.168.10.1/24 — Corporate LAN
- **Port 3 (Guest):** 192.168.20.1/24 — Isolated Guest segment
- Default route configured through the WAN gateway for outbound traffic.

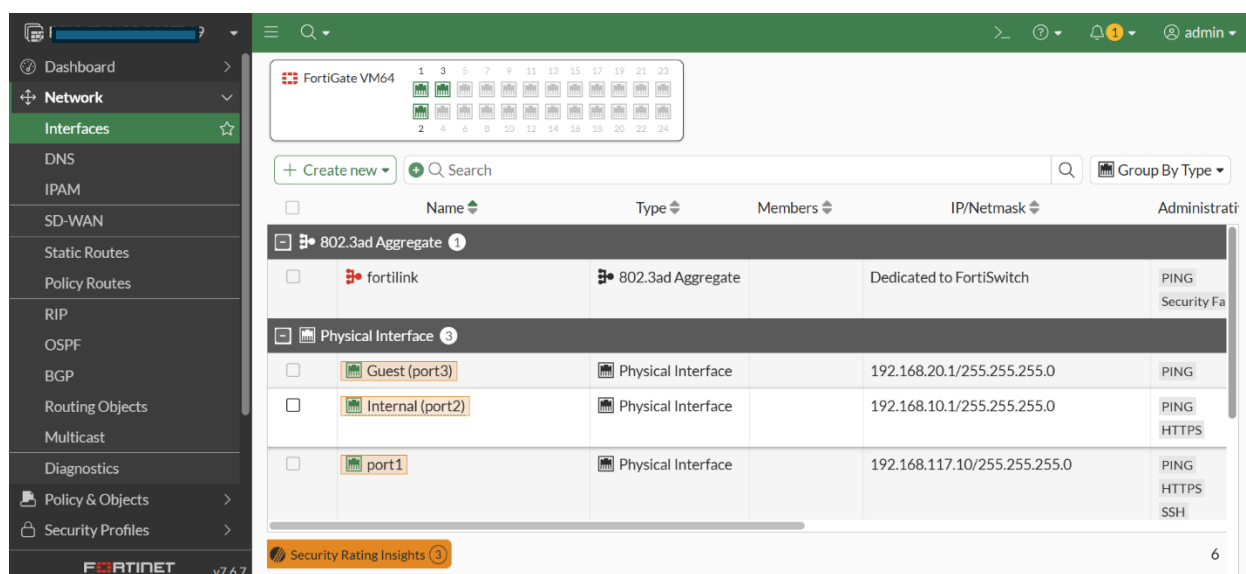


Figure 1: Network Interfaces

2. DHCP & DNS Services

Configured the firewall to handle addressing and name resolution for both internal segments.

- DHCP scopes: 192.168.10.10–100 for Internal and 192.168.20.10–100 for Guest, each with its own gateway and DNS settings.
- DNS enabled in recursive mode on both interfaces, so clients resolve through the firewall instead of going out directly.

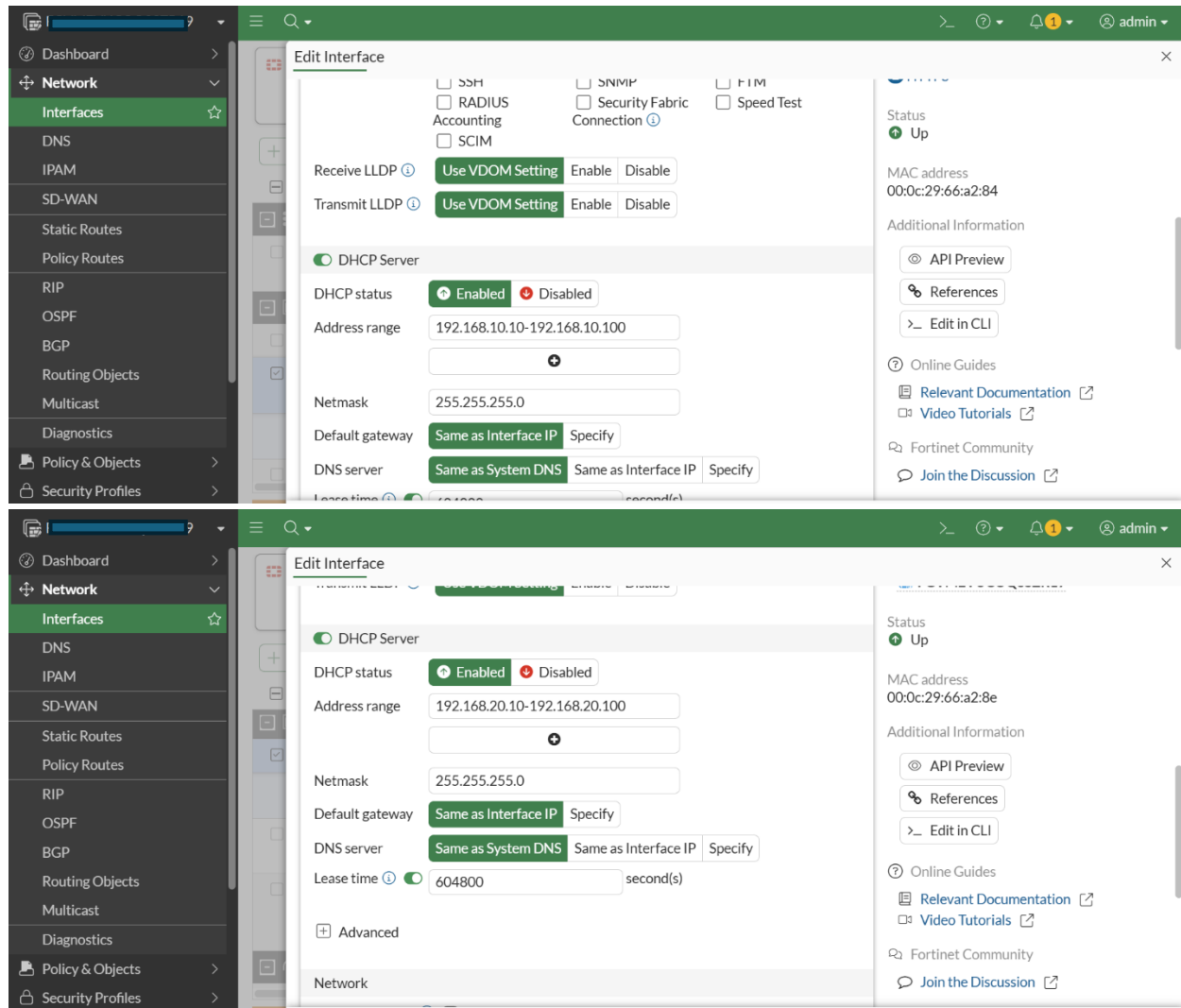


Figure 2: DHCP Server Configuration — Internal and Guest segments

3. Firewall Policies & NAT

Defined firewall rules to control traffic flow and isolate sensitive network segments:

- **Guest Isolation:** Explicitly blocked traffic from Guest (Port 3) to Internal (Port 2) to enforce network segmentation and least-privilege access between zones.
- **Internet Access:** Enabled outbound internet access for both Internal and Guest networks with NAT applied on Port 1.
- **Logging:** Enabled on the deny policy to record every blocked attempt between zones.

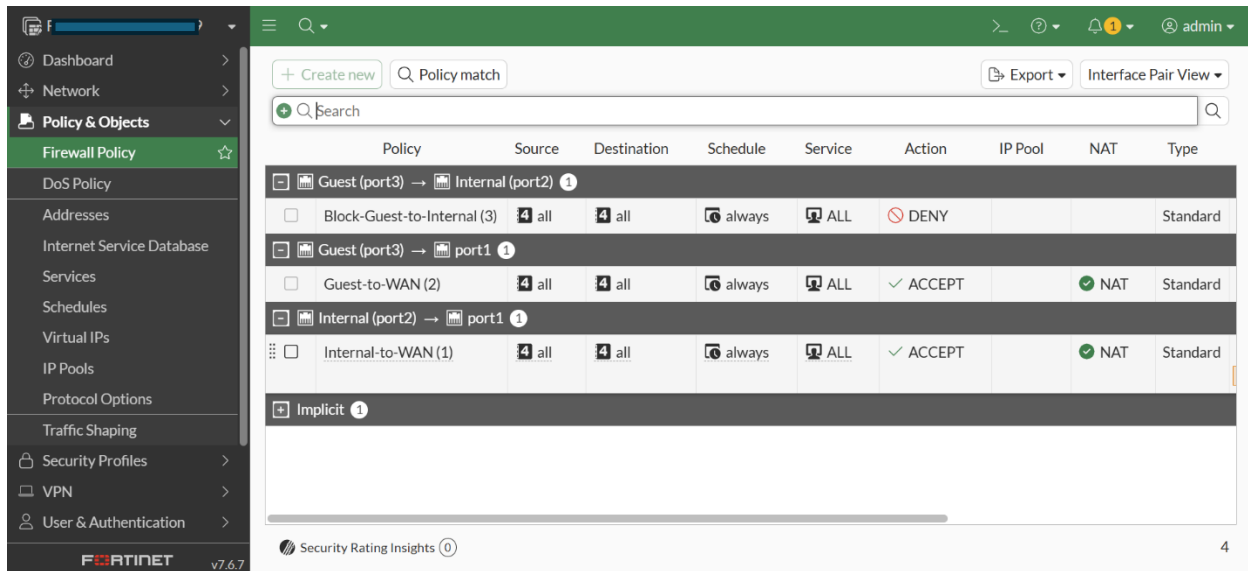


Figure 3: Firewall Policies

4. Web Filtering

Created a custom security profile (Internal-WebFilter) paired with Certificate Inspection and linked directly to the primary outbound policy (Internal-to-WAN).

- *Certificate inspection applied so filtering also covers HTTPS traffic.*
- *Installed the firewall's CA certificate on the client to support inspection without browser trust errors.*

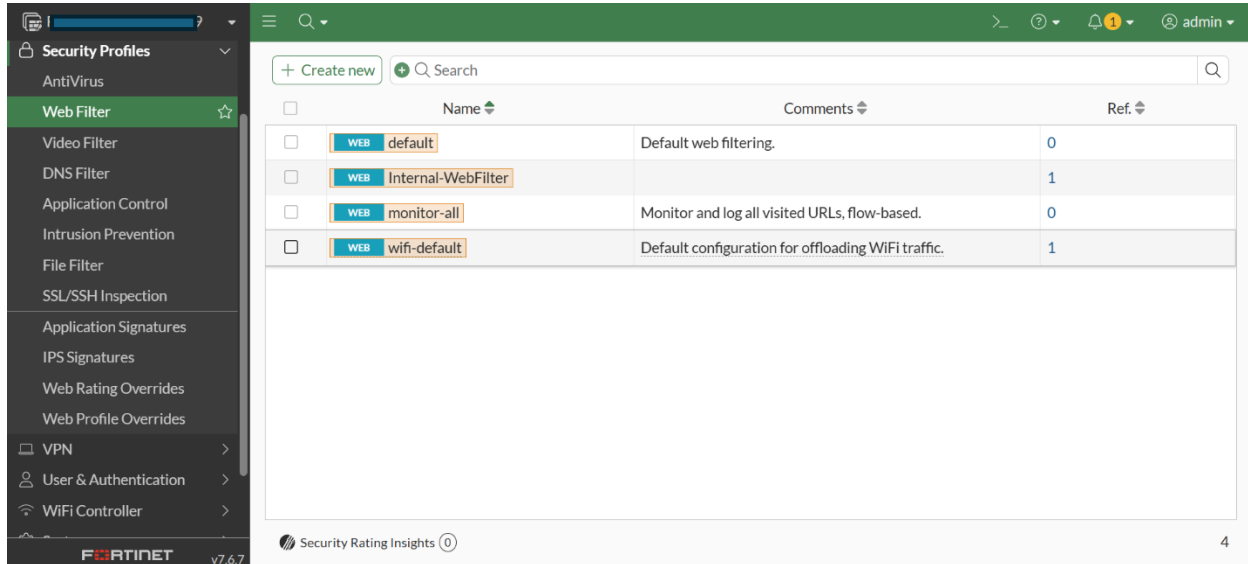


Figure 4: Web Filter Profile Overview

5. URL Blocking

Configured URL filtering rules using Wildcard patterns (facebook.com, instagram.com, gambling domains) set to Block to enforce corporate browsing compliance.

FortiGuard category-based filtering was unavailable on the evaluation license, so static URL filtering with wildcard matching was used to achieve the same enforcement.

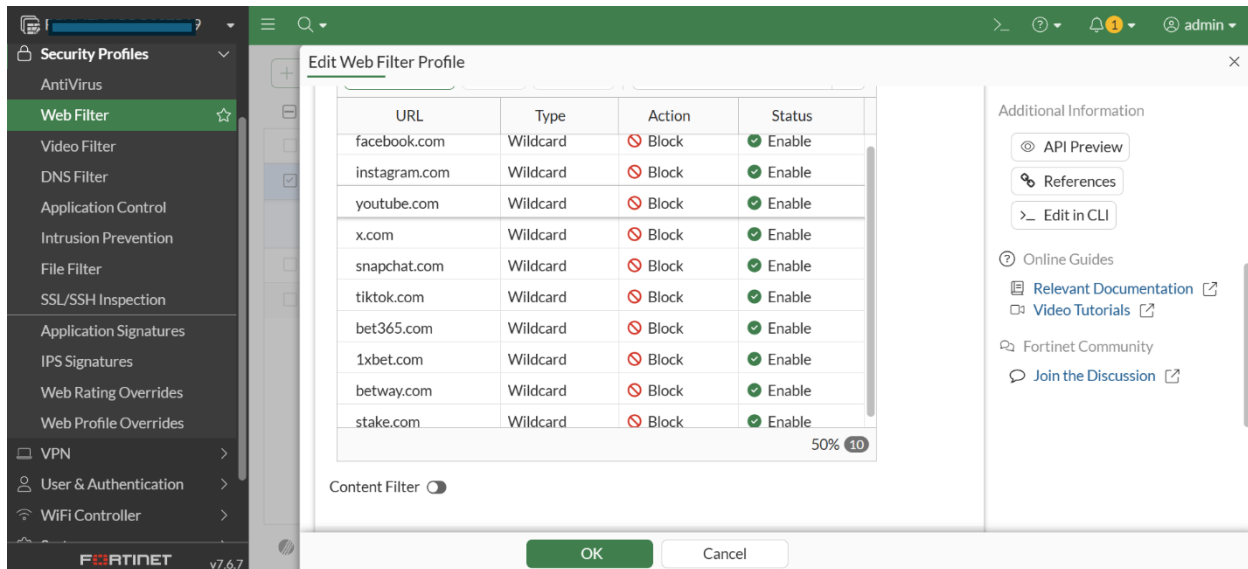
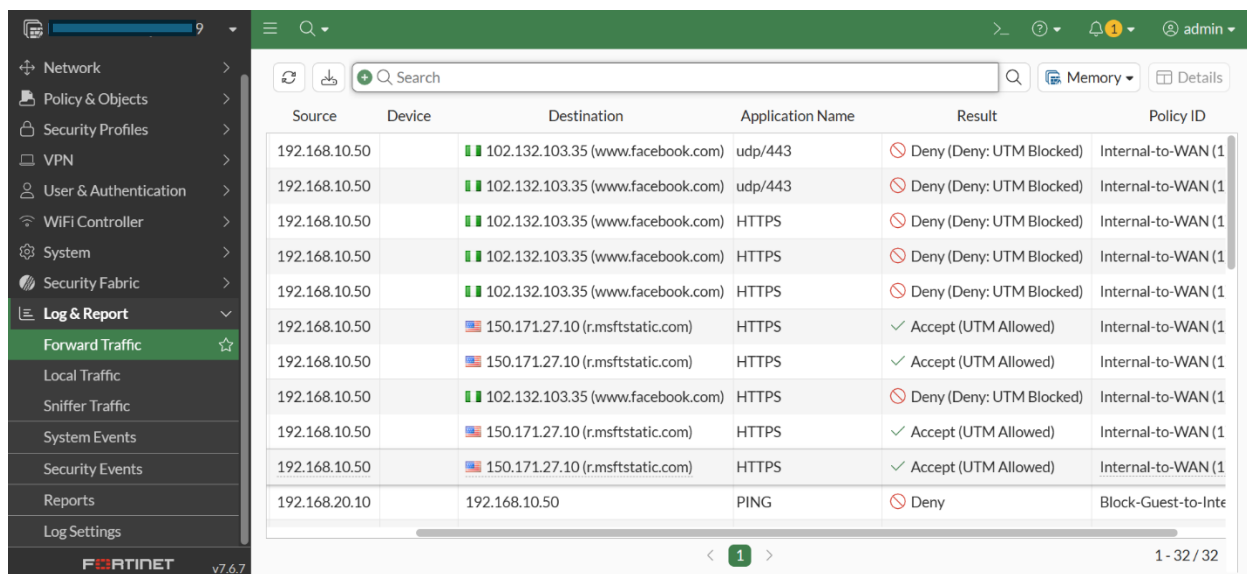


Figure 5: URL Block List Details

6. Verification & Testing

Analyzed real-time Forward Traffic logs to verify policy execution:

- **Web Block Verification:** Blocked social media requests logged as Deny (Deny: UTM Blocked).
- **Inter-Zone Isolation:** Blocked PING traffic from Guest (192.168.20.10) to Internal (192.168.10.50) logged as Deny.
- **Client-Side Verification:** Confirmed internet access, DHCP leasing, and DNS resolution from both the Windows and Ubuntu clients.



Source	Device	Destination	Application Name	Result	Policy ID
192.168.10.50		102.132.103.35 (www.facebook.com)	udp/443	Deny (Deny: UTM Blocked)	Internal-to-WAN (1
192.168.10.50		102.132.103.35 (www.facebook.com)	udp/443	Deny (Deny: UTM Blocked)	Internal-to-WAN (1
192.168.10.50		102.132.103.35 (www.facebook.com)	HTTPS	Deny (Deny: UTM Blocked)	Internal-to-WAN (1
192.168.10.50		102.132.103.35 (www.facebook.com)	HTTPS	Deny (Deny: UTM Blocked)	Internal-to-WAN (1
192.168.10.50		102.132.103.35 (www.facebook.com)	HTTPS	Deny (Deny: UTM Blocked)	Internal-to-WAN (1
192.168.10.50		150.171.27.10 (r.msftstatic.com)	HTTPS	Accept (UTM Allowed)	Internal-to-WAN (1
192.168.10.50		150.171.27.10 (r.msftstatic.com)	HTTPS	Accept (UTM Allowed)	Internal-to-WAN (1
192.168.10.50		102.132.103.35 (www.facebook.com)	HTTPS	Deny (Deny: UTM Blocked)	Internal-to-WAN (1
192.168.10.50		150.171.27.10 (r.msftstatic.com)	HTTPS	Accept (UTM Allowed)	Internal-to-WAN (1
192.168.10.50		150.171.27.10 (r.msftstatic.com)	HTTPS	Accept (UTM Allowed)	Internal-to-WAN (1
192.168.20.10		192.168.10.50	PING	Deny	Block-Guest-to-Inte

Figure 6: Forward Traffic Logs